

The Audit Chain as a Gaming Defense: Provenance, Reproducibility, and Tamper Detection

R.4 — Adversarial Robustness Track

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Abstract

The BISECT redistricting platform’s principal defense against gaming is a cryptographic audit chain linking input data, computation, and output through SHA-256 hashes. This paper provides a formal analysis of the audit chain as a gaming defense: its cryptographic security properties, its three-level provenance architecture, and its fitness for use as litigation-grade reproducibility evidence under the Daubert standard.

The SHA-256 collision resistance bound (2^{80} operations under the birthday paradox) makes undetected manifest manipulation computationally infeasible. At 10^{15} operations per second on 2026 hardware, generating a SHA-256 collision requires approximately 3×10^7 years.

We demonstrate `bisect label-verify` on three contested states: North Carolina (2020), Texas (2020), and Wisconsin (2020). For each state, we show that any party with access to the public manifest can independently verify the full computation chain from Census Bureau download through final plan assignment in under 60 seconds on a standard workstation.

Under Daubert’s four criteria (testability, peer review, error rate, general acceptance), the BISECT audit chain satisfies all four: it is testable by any party, the underlying

SHA-256 algorithm is peer-reviewed and standardized (FIPS PUB 180-4), the error rate (false non-detection) is bounded by 2^{-80} , and SHA-256 is universally accepted in cryptographic practice.

We provide model DISTRICTING INTEGRITY ACT legal language for SHA-256 manifest filing: the manifest must be filed with the Secretary of State within 30 days of plan adoption and retained for 10 years. This language is designed to be directly adoptable by state legislatures considering algorithmic redistricting statutes.

Keywords

SHA-256, audit chain, provenance, reproducibility, Daubert standard, tamper detection, FIPS 180-4, bisect label-verify, manifest, DIA

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1. Introduction

The Track R program (R.0–R.3) has established that all gaming vectors available to a hostile actor deploying the BISECT redistricting platform are either bounded below the noise floor (R.1: parameter gaming, ≤ 0.3 seats; R.2: input data manipulation, ≤ 0.4 seats) or pre-specified by statute (R.3: geographic definitions).

But bounding the maximum effect of a gaming attack is not sufficient for legal purposes. A court needs to know: given this specific plan, can we verify that no gaming occurred? This is a question of *provenance*—the ability to trace the plan from its inputs to its outputs and verify that the computation was performed as claimed.

Provenance has a long history in forensic science and litigation. The chain of custody for physical evidence establishes provenance: who had the evidence, when, and what they did with it. For algorithmic redistricting, the analogous chain of custody is cryptographic: SHA-256 hashes link the Census Bureau’s public data releases through the computation to the final plan, with each link in the chain independently verifiable.

This paper formalises the BISECT audit chain as a legal defense. We address three questions:

1. **Cryptographic:** How strong is the SHA-256 guarantee? What is the probability that an adversary can tamper with any link in the chain without detection? (Section 2)
2. **Technical:** How does the three-level provenance architecture work in practice, and how does `bisect label-verify` demonstrate it for a specific plan? (Sections 3–4)
3. **Legal:** Does the audit chain satisfy the Daubert standard for admissibility of scientific evidence? What statutory language should a state legislature adopt to make the manifest filing enforceable? (Sections 5–6)

1.1. Why Reproducibility Matters in Redistricting Litigation

Expert witnesses in redistricting cases routinely present computational analyses to courts. The opposing party can challenge such analyses on the grounds that the expert’s results are not reproducible: that running the same algorithm on the same data would not produce the same results, or that the expert has not disclosed the inputs and parameters used.

The BISECT audit chain addresses this challenge directly. The manifest contains everything needed to reproduce the analysis: algorithm version hash, parameter configuration, data download hashes, adjacency construction hash, and output plan hash. Any qualified expert can run `bisect label-verify` and confirm— or refute—the original expert’s claim.

This converts redistricting expert testimony from a credibility contest (whose algorithm is more trustworthy?) into a verifiable computation (does the manifest check out?).

2. SHA-256 Cryptographic Analysis

2.1. Algorithm Specification

SHA-256 is a member of the SHA-2 family of cryptographic hash functions, standardised by NIST in FIPS PUB 180-4 [National Institute of Standards and Technology, 2015]. It maps an arbitrary-length input to a 256-bit (32-byte) output.

Formally, SHA-256 satisfies three properties [Bellare and Kohno, 2004]:

1. **Pre-image resistance:** Given a hash output h , it is computationally infeasible to find an input m such that $\text{SHA256}(m) = h$.
2. **Second pre-image resistance:** Given an input m_1 , it is computationally infeasible to find a different input $m_2 \neq m_1$ such that $\text{SHA256}(m_1) = \text{SHA256}(m_2)$.
3. **Collision resistance:** It is computationally infeasible to find any two distinct inputs $m_1 \neq m_2$ such that $\text{SHA256}(m_1) = \text{SHA256}(m_2)$.

2.2. The Birthday Bound

By the birthday paradox, a collision attack on a hash function with n -bit output requires approximately $2^{n/2}$ hash evaluations in expectation. For SHA-256, this is 2^{128} evaluations.

However, the relevant security property for the audit chain is second pre-image resistance (Property 2), not collision resistance. The adversary has a specific target: the legitimate file with its known hash. They need to find a *different* file that hashes to the same value. Second pre-image attacks are harder than collision attacks; the best known bound is 2^{256} for pre-image and 2^{128} for second pre-image under the generic attack model.

For audit chain purposes (does this file match this hash?), we use the collision resistance bound as a conservative estimate: 2^{128} operations. Under the birthday bound interpretation (finding any collision among 2^{80} attempted values with 50% probability), the bound is 2^{80} .

Theorem 1 (SHA-256 Audit Chain Security). *Under SHA-256, an adversary cannot produce a modified input file that has the same SHA-256 hash as the original without performing at least 2^{80} hash evaluations (birthday bound for collision finding). At 10^{15} evaluations per second on 2026-grade hardware, this requires approximately 3×10^7 years.*

2.3. Known Attack Status

As of 2026, no practical attacks on SHA-256 are known. The best theoretical attacks (differential cryptanalysis, reduced-round attacks) apply only to reduced versions of SHA-256 and do not reduce the security below 2^{128} [Mendel et al., 2013]. SHA-256 is approved for use in U.S. government applications under NIST SP 800-57 through at least 2030.

2.4. Quantum Resistance

Grover’s algorithm [Grover, 1996] provides a quadratic speedup for pre-image attacks on hash functions using a quantum computer, reducing the effective security from 2^{256} to 2^{128} . For the collision resistance bound relevant to the audit chain, Grover’s algorithm provides no improvement (collision-finding does not reduce below 2^{128} even with quantum speedup).

The BISECT audit chain is therefore quantum-resistant at the 2^{128} security level for the collision-finding task.

3. Three-Level Provenance Architecture

3.1. Architecture Overview

The BISECT audit chain comprises three levels, each hashing the output of the previous level and the inputs at that level.

Definition 1 (bisect Manifest). *A BISECT manifest $\mathcal{M}$ is a structured JSON document*

containing:

$$\begin{aligned}\mathcal{M} = \{ & \textit{algorithm_hash} : h_{\text{alg}}, \\ & \textit{config_hash} : h_{\text{cfg}}, \\ & \textit{download_hashes} : \{h_{\text{pl94171}}, h_{\text{tiger}}, h_{\text{acs}}\}, \\ & \textit{adjacency_hash} : h_{\text{adj}}, \\ & \textit{plan_hash} : h_{\text{plan}}, \\ & \textit{manifest_hash} : h_{\text{manifest}} \}\end{aligned}$$

where $h_{\text{manifest}} = \text{SHA256}(\mathcal{M} \setminus \{h_{\text{manifest}}\})$ is a self-referential integrity hash.

3.2. Level 1: Data Provenance

Level 1 establishes that the algorithm used the Census Bureau’s published data and not a modified version. The three Level 1 hashes are:

- $h_{\text{pl94171}} = \text{SHA256}(\text{P.L. 94-171 file})$: verified against Census Bureau’s published checksum at <https://www2.census.gov/>.
- $h_{\text{tiger}} = \text{SHA256}(\text{TIGER/Line shapefile ZIP})$: verified against Census Bureau’s TIGER archive.
- $h_{\text{acs}} = \text{SHA256}(\text{ACS 5-year table CSV})$: verified against Census Bureau’s ACS data portal.

Any party can independently verify these hashes by downloading the same files from the Census Bureau and computing SHA-256.

3.3. Level 2: Computation Provenance

Level 2 establishes that the adjacency graph was correctly constructed from the Level 1 data.

$$h_{\text{adj}} = \text{SHA256}(G(h_{\text{tiger}}))$$

where $G(h_{\text{tiger}})$ denotes the adjacency graph constructed from the TIGER shapefile identified by h_{tiger} .

Level 2 is verified by: (a) independently downloading the TIGER shapefile and verifying h_{tiger} , then (b) running BISECT’s adjacency construction and checking that it produces h_{adj} .

3.4. Level 3: Output Provenance

Level 3 establishes that the output plan was produced by running the specified algorithm binary with the specified parameters on the specified data.

$$h_{\text{plan}} = \text{SHA256}(\text{bisect}(h_{\text{alg}}, h_{\text{cfg}}, h_{\text{pl94171}}, h_{\text{adj}}))$$

Level 3 is verified by `bisect label-verify`, which replays the full computation from the manifest and checks that the output hash matches h_{plan} .

3.5. Chain Integrity

The manifest hash h_{manifest} binds all six hashes together. Any modification to any level—algorithm binary, configuration, download, adjacency, plan, or the manifest itself—changes h_{manifest} . A modified manifest is detectable by anyone with the original manifest hash, which is filed with the Secretary of State under the DISTRICTING INTEGRITY ACT protocol.

Theorem 2 (Manifest Integrity). *Given the SHA-256 collision resistance bound, an adversary cannot produce a modified manifest $\mathcal{M}'$ such that: (a) $\mathcal{M}' \neq \mathcal{M}$, and (b) $h_{\text{manifest}}(\mathcal{M}') = h_{\text{manifest}}(\mathcal{M})$, without performing $\geq 2^{80}$ hash evaluations.*

4. Demonstration: `bisect label-verify` for NC, TX, WI

We demonstrate `bisect label-verify` for three states selected to represent the range of congressional delegation sizes and litigation history.

4.1. North Carolina (k=14)

North Carolina is the most frequently litigated redistricting state in the United States since 2010. Multiple plans have been challenged in both federal and state courts. The BISECT NC 2020 plan provides a comparison point.

```
$ bisect label-verify nc_official_2020 --year 2020
[INFO] Loading manifest: nc_official_2020_2020.manifest.json
```



```

[INFO] Algorithm binary hash: b4a7f3...
      Expected (registered): b4a7f3... MATCH
[INFO] Config hash: 9c1d82...
      Expected (registered): 9c1d82... MATCH
[INFO] PL94-171 download hash: a7f3bc...
      Census Bureau published: a7f3bc... MATCH
[INFO] TIGER shapefile hash: d4e82c...
      Census Bureau published: d4e82c... MATCH
[INFO] ACS table hash: f1c0a9...
      Census Bureau published: f1c0a9... MATCH
[INFO] Adjacency graph hash: 8b2a7d...
      Recomputing adjacency from downloaded shapefile...
      Computed: 8b2a7d... MATCH
[INFO] Plan hash: 2e9f4c...
      Replaying computation from manifest...
      Computed plan hash: 2e9f4c... MATCH
[PASS] Full audit chain verified for NC 2020
[INFO] Verification time: 43 seconds

```

4.2. Texas (k=38)

Texas has the second-largest congressional delegation and has undergone mid-decade redistricting.

```

$ bisect label-verify tx_official_2020 --year 2020
[INFO] Loading manifest: tx_official_2020_2020.manifest.json
[INFO] Algorithm binary hash: b4a7f3... MATCH
[INFO] Config hash: 9c1d82... MATCH
[INFO] PL94-171 download hash: c2e8a1... MATCH
[INFO] TIGER shapefile hash: 7f3d19... MATCH
[INFO] ACS table hash: e4b2c8... MATCH
[INFO] Adjacency graph hash: 1a7c4f...
      Recomputing adjacency (38-seat state; larger graph)...
      Computed: 1a7c4f... MATCH
[INFO] Plan hash: 5d8e2b...

```

```
Replaying ConvergenceSweep T=600 (estimated 12 sec)...
Computed plan hash: 5d8e2b... MATCH
[PASS] Full audit chain verified for TX 2020
[INFO] Verification time: 57 seconds
```

4.3. Wisconsin ($k=8$)

Wisconsin is the prototypical packed/cracked gerrymander from B.0’s algorithm bakeoff. The comparison between the enacted partisan gerrymander and the BISECT algorithmic plan is documented in G.1.

```
$ bisect label-verify wi_official_2020 --year 2020
[INFO] Loading manifest: wi_official_2020_2020.manifest.json
[INFO] Algorithm binary hash: b4a7f3... MATCH
[INFO] Config hash: 9c1d82... MATCH
[INFO] PL94-171 download hash: 3b9e5f... MATCH
[INFO] TIGER shapefile hash: 6c1a4d... MATCH
[INFO] ACS table hash: 8e7b3a... MATCH
[INFO] Adjacency graph hash: 4d2c9e... MATCH
[INFO] Plan hash: 7a1b5f...
Replaying ConvergenceSweep T=600 (WI, 8 districts)...
Computed plan hash: 7a1b5f... MATCH
[PASS] Full audit chain verified for WI 2020
[INFO] Verification time: 31 seconds
```

4.4. Summary

All three states verify in under 60 seconds on a standard 2024-generation workstation (8-core Intel Core i7, 16 GB RAM). The computation time scales with delegation size (k): WI ($k = 8$, 31 sec), NC ($k = 14$, 43 sec), TX ($k = 38$, 57 sec). For California ($k = 52$), expected verification time is approximately 90 seconds.

The verification process is fully deterministic: running `bisect label-verify` twice on the same manifest produces identical results. This determinism is the foundation of the Daubert argument in Section 5.

5. Daubert Standard Analysis

5.1. The Daubert Framework

In *Daubert v. Merrell Dow Pharmaceuticals* [dau, 1993], the Supreme Court established four criteria for admissibility of scientific expert testimony in federal courts:

1. **Testability:** The theory or technique can be and has been tested.
2. **Peer review and publication:** The theory or technique has been subjected to peer review and publication.
3. **Known or potential error rate:** The known or potential error rate of the technique is acceptable.
4. **General acceptance:** The theory or technique is generally accepted in the relevant scientific community.

Daubert applies directly to federal proceedings; most states have adopted similar standards.

5.2. The Audit Chain Under Daubert

Criterion 1: Testability. The BISECT audit chain is maximally testable: any party with access to the public manifest and an internet connection can run `bisect label-verify` and independently confirm or refute the original expert’s computation. This exceeds the typical testability requirement: most scientific techniques require specialized equipment or training to test, whereas `bisect label-verify` can be run by any qualified expert in redistricting or computer science. **Criterion 1: Satisfied.**

Criterion 2: Peer review and publication. The SHA-256 algorithm is standardized by NIST in FIPS PUB 180-4 [National Institute of Standards and Technology, 2015] and has been subjected to extensive peer review and cryptanalysis over two decades. The BISECT algorithm is documented in B.1 [Della-Libera, 2026] and subsequent papers, and the audit chain architecture is described in this paper. **Criterion 2: Satisfied.**

Criterion 3: Known error rate. The false non-detection rate of the SHA-256 audit chain is bounded by 2^{-80} (probability that a tampered manifest passes verification under the birthday attack model). This error rate is 10^{-24} , far below any reasonable threshold for legal evidence (which typically requires only that evidence be “more probable than not” or, for criminal cases, “beyond reasonable doubt”). **Criterion 3: Satisfied.**

Criterion 4: General acceptance. SHA-256 is the most widely deployed cryptographic hash function in the world, used in TLS/SSL, code signing, version control (git), and government document integrity verification. It is accepted by NIST, FIPS, NSA, and the cryptographic research community. **Criterion 4: Satisfied.**

5.3. The Reproducibility Advantage

Beyond satisfying Daubert’s four criteria, the audit chain provides a reproducibility guarantee that is unusual in redistricting expert testimony. Typically, an expert’s computational analysis is reproducible in principle but impractical in practice: the opposing party would need to obtain the same software, same data, and same parameter settings, and then run the same analysis.

With BISECT’s audit chain, reproducibility is:

- **Automatic:** `bisect label-verify` performs the reproduction.
- **Fast:** Under 60 seconds for any state (see Section 4).
- **Exact:** The reproduction is bit-for-bit identical; there is no “reproduction with minor variations.”
- **Open:** The BISECT binary is open-source; the algorithm and its implementation can be audited by any qualified expert.

This makes the expert’s computation “reproducibly falsifiable”—the opponent does not need to dispute the expert’s methodology; they need only run `bisect label-verify` and check whether it passes or fails.

6. Model DIA Legal Language

6.1. Purpose

This section provides model statutory language for state legislatures considering the DISTRICTING INTEGRITY ACT. The language is designed to be directly adoptable and covers three elements: (1) mandatory manifest filing with the Secretary of State, (2) retention period, and (3) public verification rights.

6.2. Model Statutory Language

Section [X]. Algorithmic Redistricting Manifest.

(a) *Manifest generation.* Any redistricting plan adopted pursuant to this Act shall be accompanied by a cryptographic computation manifest (“Manifest”) generated by the certified redistricting algorithm. The Manifest shall contain, at minimum:

1. The SHA-256 hash of the redistricting algorithm binary used to generate the plan, identifying the specific version of the algorithm;
2. The SHA-256 hash of the parameter configuration file used to generate the plan;
3. The SHA-256 hash of each Census Bureau data file used as input to the algorithm, as provided by the U.S. Census Bureau’s official redistricting data release;
4. The SHA-256 hash of the adjacency graph constructed from Census Bureau TIGER/Line shapefiles;
5. The SHA-256 hash of the final district assignment output;
6. A self-referential integrity hash computed as the SHA-256 hash of the foregoing contents of the Manifest.

(b) *Filing requirement.* The redistricting authority shall file the Manifest with the Secretary of State within thirty (30) days of plan adoption. The Secretary of State shall make the Manifest available for public inspection on the Secretary of State’s official website within five (5) business days of receipt.

(c) *Retention period.* The redistricting authority shall retain all source data, parameter configuration files, algorithm binary, and intermediate computation outputs referenced in the Manifest for a period of not less than ten (10) years following the effective date of the redistricting plan, or until such time as any pending litigation arising from the plan is finally resolved, whichever is later.

(d) *Verification rights.* Any registered voter in the state or any candidate who has filed for office in the state shall have the right to:

1. Obtain a copy of the Manifest from the Secretary of State;
2. Independently verify the Manifest by running the certified redistricting algorithm's verification tool against the Manifest;
3. Request from the redistricting authority, at no charge, a copy of any source data or parameter file referenced in the Manifest.

(e) *Violation.* A redistricting plan whose Manifest fails verification under the certified algorithm's verification tool shall be voidable at the instance of any person who has standing to challenge the plan. The redistricting authority bears the burden of demonstrating that a verification failure is attributable to technical error rather than data manipulation.

(f) *Expert testimony.* In any proceeding challenging a redistricting plan adopted under this Act, the Manifest and the results of the verification tool constitute admissible evidence of the computation's provenance and integrity, subject to the provisions of the state rules of evidence governing scientific and technical evidence.

6.3. Commentary

The 30-day filing deadline gives the redistricting authority time to complete verification but prevents indefinite delay. The 10-year retention period covers the full decade until the next redistricting cycle plus a buffer for litigation that may extend beyond the cycle. The verification right for any registered voter reflects the public nature of redistricting and ensures that opponents do not need to file formal legal proceedings to access verification tools.

The burden-shifting in subsection (e) reflects the technical reality that SHA-256 verification failures are almost never technical errors; the probability of a false failure under correct

computation is negligible. Requiring the redistricting authority to explain a failure creates an accountability mechanism without creating a presumption of fraud for genuinely technical edge cases.

7. Conclusion

The BISECT audit chain—three levels of SHA-256 hashes linking Census Bureau data through computation to final plan—is a cryptographically sound defense against all gaming vectors identified in Track R.

The security argument is simple: SHA-256 collision resistance makes undetected manifest manipulation computationally infeasible at 2^{80} operations. Any gaming attack that modifies input data, algorithm parameters, or computation outputs changes at least one hash in the chain. That change is detectable by any party who runs `bisect label-verify`.

The Daubert analysis confirms that this audit chain meets all four criteria for admissible scientific evidence: testability, peer review, bounded error rate (2^{-80}), and general acceptance of SHA-256. This makes the audit chain not merely a technical safeguard but a litigation-grade provenance instrument.

The model DIA statutory language in Section 6 translates the technical audit chain into enforceable legal obligations: 30-day manifest filing, 10-year retention, public verification rights, and a burden-shifting rule for verification failures. State legislatures can adopt this language directly.

The Track R program’s conclusion follows from R.0–R.4 in combination: when the DIA protocol is followed, the maximum achievable partisan shift from all gaming vectors combined is ≤ 0.5 Democratic seats nationally (within the algorithmic noise floor), and any larger manipulation is detectable with probability approaching 1. Algorithmic redistricting under the DIA protocol is not merely partisan-blind; it is verifiably partisan-blind.

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